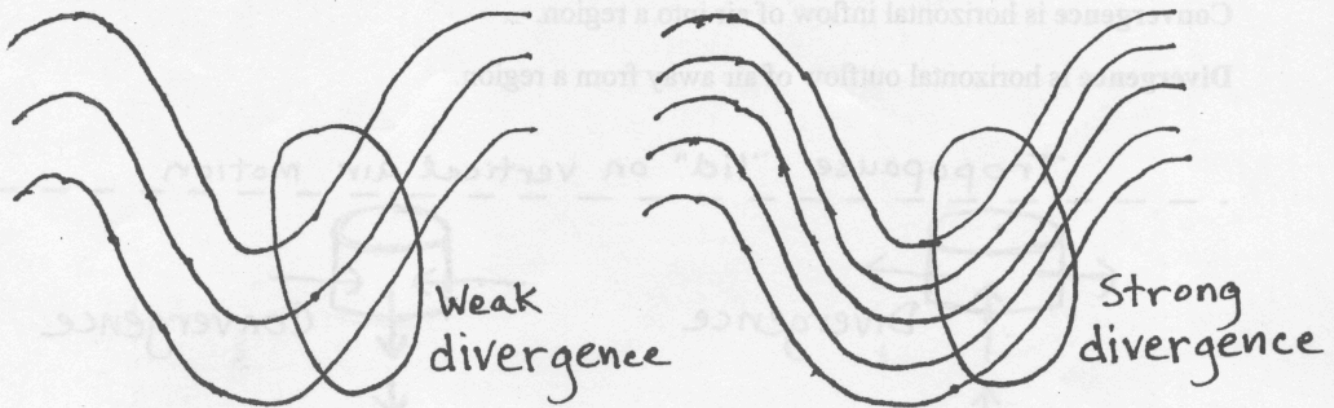


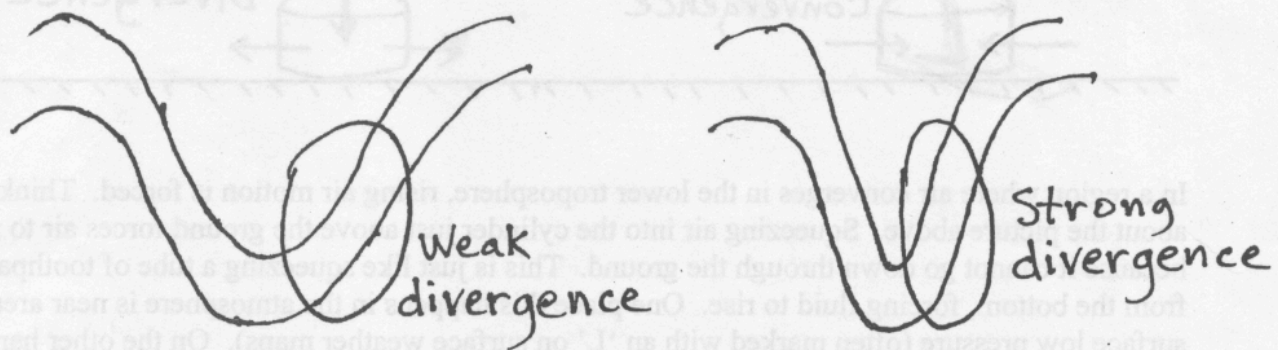


The shape of a 500 mb trough often indicates something about its dynamical strength, i.e., its potential to force strong rising motion in the atmosphere and hence strong areas of precipitation. This page describes some things to look for in the 500 mb pattern that can help to identify strong weather systems that will have strong upper tropospheric divergence and forced rising air.

1. Stronger 500 mb winds increase upper tropospheric divergence.



2. The more amplified the pattern, the stronger the upper tropospheric divergence. A picture is provided in the class lecture notes titled "The 500 mb height pattern and surface weather."
3. The sharper the curvature of a trough, the stronger the divergence.



4. A "negative tilt" trough often indicates a potent and intensifying system, while a "positive tilt" trough is usually a weaker system. Negative tilt troughs tilt from northeast toward southwest and positive tilt troughs tilt from northwest toward southeast.

